Tejessrihari Prabakaran

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Education

Georgia Institute of Technology — Graduation Date: May 2027
BS in Computer Science — **GPA: 3.7/4.0**

- **Specialization:** Artificial Intelligence & Modeling and Simulation
- **Coursework:** Data Structures, Design and Analysis of Algorithms, Machine Learning, Systems & Networks, Deep Learning, Computer Vision

Experience

Software Engineer Intern, The Home Depot – Smyrna, GA — May 2026 – Jul 2026

- Built a Gemini 3.1 Pro graph-traversal system replacing flat-table lookups with graph-based reasoning for developer queries, saving ∼**5,000 developer-hours annually** via a 72.2% to 88.0% (880/1,000 correct) accuracy lift, outperforming the legacy internal Q&A tool by 14.7 pts.
- Prevented prompt/config regressions from ever shipping, validated across **2,000 augmented questions**, by gating every change behind a held-out accuracy check in a 4-phase, 15-iteration self-tuning pipeline.
- Eliminated manual redeploys across the graph-traversal system, tuning pipeline, and chatbot by automating deployment and retraining – containerizing all three on auto-scaling Cloud Run via GitHub Actions CI/CD with a **6h cron-gated retrain cycle**.
- Gave previously siloed teams shared lineage traceability across **6,534 assets** and 64M lineage entries, by modeling ownership and dependencies as a BigQuery property graph spanning metadata, lineage, and semantic subgraphs.
- Validated the graph-traversal system's reliability, not just a single accuracy snapshot, by measuring **8x tighter output variance** than the flat-table baseline.

Software Engineer Intern, The Home Depot – Smyrna, GA — May 2025 – Jul 2025

- Designed and shipped an incident-debugging API with automated multi-step workflows on GCP that auto-diagnoses production issues, improving developer **productivity 40%** and shrinking **response time 38%** (2.6h to 1.6h) across ∼30 incidents/month.
- Automated root-cause classification for production incidents, achieving an **F1 of 0.86** across 9 error classes including complex cross-stack function-call errors, by building a Cloud Logging-to-BigQuery pipeline and training scikit-learn classifiers exposed via REST endpoints.
- Reduced **duplicate tickets 22%** and **IDE lookup time by 41s** by orchestrating a Google-ADK pipeline of Gemini agents that rank candidate fixes using BM25.
- Drove adoption across **100+ developers** for faster in-IDE debugging by integrating the debugging API via MCP directly into IDEs to fetch logs and traces.
- Surfaced root-cause signals from four separate observability layers into one unified workflow, by instrumenting the debugging API across BigQuery, Cloud Tracing, Cloud Logging, and Kubernetes.

Projects

Autonomous SOC Platform *(Kafka, Beam, Elasticsearch, XGBoost, eBPF, Gemini 3.1 Pro)* — Apr 2026 – Current

- Built an autonomous SOC platform that triages security incidents in real time, compressing **triage time 45min to 3min**, reducing hallucination 23% to 4%, and curbing false-positive escalations 67% via a RAG-grounded Gemini 3.1 Pro copilot across 3.8M events.
- Automated malicious-vs-benign event triage, achieving **ROC-AUC 0.975** (Precision 0.901, Recall 0.875), by training XGBoost classifiers on 3.8M honeypot events with BorderlineSMOTE and Optuna Bayesian tuning.
- Sustained **500K events/sec** ingestion at <1% CPU overhead by streaming 3.8M events through a partition-keyed Kafka/Apache Beam pipeline and a least-privilege eBPF DaemonSet (no root) for kernel telemetry.
- Delivered **sub-20ms semantic search** via dual BM25/HNSW retrieval on Elasticsearch (384-dim embeddings), deployed on Kubernetes with horizontal pod autoscaling and rolling updates.

Hybrid LLM Processor *(FastAPI, Presidio/spaCy, scikit-learn, Ollama/Qwen3, Gemini, ReactJS, Docker)* — May 2025 – Dec 2025

- Built a hybrid local/cloud LLM system routing each request between a self-hosted Qwen3 14B model and Gemini via a scikit-learn TF-IDF classifier (**0.95 CV accuracy**), keeping routine queries local and reserving cloud calls for hard cases to balance cost, latency, and privacy.
- Kept sensitive data structurally unreachable by the cloud path – **zero leaks** across 300 concurrent sessions (4,055 requests), 0.89 routing accuracy, F1 0.921 macro – via Presidio+spaCy PII detection wired into a fail-closed routing architecture.
- Achieved **100% specificity** and zero false escalations by building a confidence-based escalation cascade that routes only weak-confidence segments to Gemini at a total added cost of $0.000319, based on Qwen3 log-probabilities.
- Decreased **cloud spend 46%** and **cloud latency 30.6%** (2105ms vs 3034ms, 172 RPS) with local Qwen3 inference, served via a Dockerized FastAPI+React app with independent circuit breakers per upstream and rate limiting.

Better Place Drones *(Python, PyTorch, Transformers, Mission Planner)* — Aug 2024 – Current

- Built a drone-based fire/object detection system, achieving **78% mAP** and 0.84 recall on 13k thermal/HD frames by training CNN + Transformer models in PyTorch with transfer learning, mixed precision, and custom-built labeled datasets.
- Optimized pipeline latency from **62ms to 11ms** by building a data pipeline for denoising, timestamp sync, and augmentation.
- Reduced mean coverage gap by **14%** in AirSim by implementing path-planning algorithms for navigation in cluttered terrain.
- Streamed live telemetry, mission uploads, and real-time overlays to operators by integrating with the Mission Planner ground station.

Technical Skills

- **Languages/Databases/Cloud:** Python, Java, C, C++, SQL, JavaScript, TypeScript, Bash, MongoDB, MySQL, PostgreSQL, Firebase, GCP
- **Frameworks/Libraries:** PyTorch, scikit-learn, Apache Beam, XGBoost, Flask, ReactJS, NumPy, LangChain, Google-ADK, Pandas
- **Architecture & Protocols:** REST APIs, Model Context Protocol (MCP), Microservices
- **DevOps/Tools:** Docker, Kubernetes, Apache Kafka, Elasticsearch, eBPF, Git, Cloud Logging, Cloud Tracing, GitHub Actions, Linux